Ocean Monitoring Indicator (OMI) of the Copernicus Marine Environment Monitoring Service

BLKSEA_OMI_HEALTH_trend

Issue: 1.3

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CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	19/02/2019	all	Creation of the document	Arthur Capet	
1.1	13/10/2020	all	Added figures	Arthur Capet	
1.2	18/08/2020	All	2020 Update	Arthur Capet	
1.3	10/06/2022	All	2022 Update	Arthur Capet	

Products name:

BLKSEA_OMI_HEALTH_OXYGEN_trend

Indicator evaluation method:

The oxygenation status of the Black Sea open basin is monitored through three complementary indicators, that are derived from vertical profiles of oxygen concentration, salinity and temperature (Capet et al, 2016).

(1) The oxygen penetration depth is defined as the depth below which oxygen concentrations are below 20 mmol/m³ (Fig. 1). This threshold is chosen to minimize the sensitivity of the indicators to the accuracy of different sensors at very low oxygen concentrations.

(2) The oxygen penetration density is defined as the potential density anomaly at the oxygen penetration depth (Fig. 2). Using pycnal vertical coordinates minimizes the uncertainty stemming from the spatial variability of oxygen penetration depth, currently undersampled by the limited number of Argo profilers. It is considered that this spatial variability, combined with the limited spatial cover of Argo sampling is the main source of error in the basin average.

(3) The oxygen inventory is the vertical integral of oxygen concentration (Fig 3.). This metric is sensitive to other aspect of the oxygen dynamics, such as decrease in the surface oxygen content due to limited oxygen solubility in warm conditions.

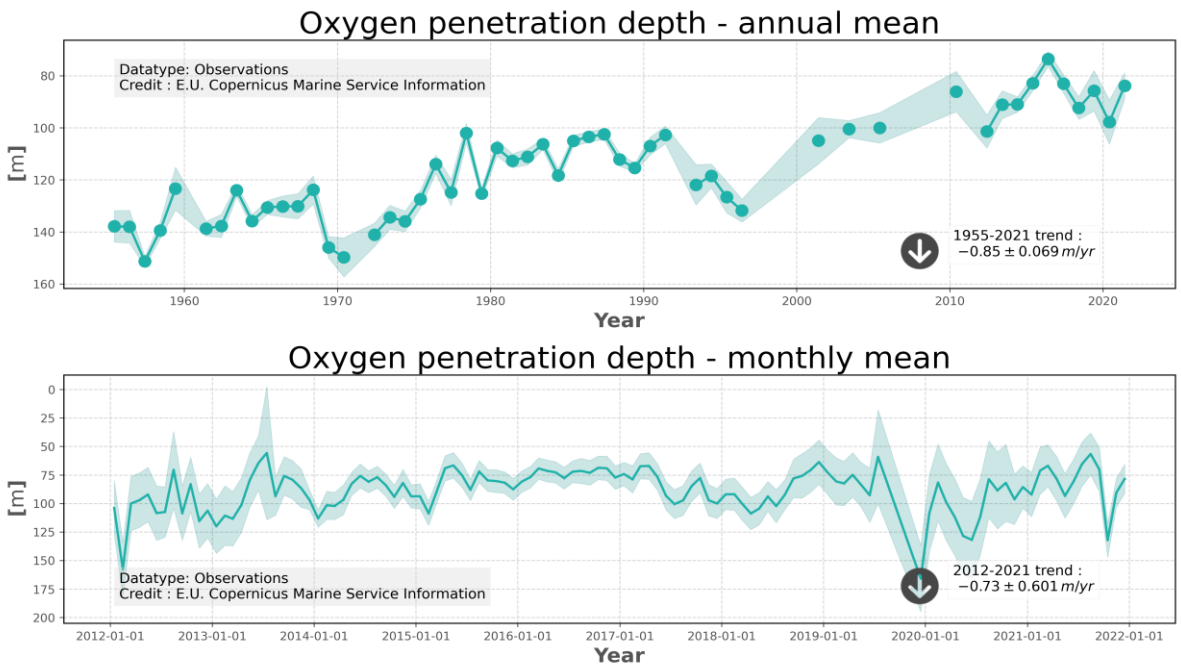

Figure 1 : Oxygen penetration depth in the open Black Sea. Upper panel: Annual trends from ship-cast (1955-2005) and autonomous platforms (Argo, 2012-2021) in-situ observations. Lower Panel: Monthly trends from Argo in-situ observations. Shaded area represent the standard error.

Each metric is extracted from individual Argo profiles, and used to compute monthly and annual averages, for the Argo Era (2012-2021). In addition, historical ship casts are gathered to extract interannual trend for each metric using the DIVA detrending protocol (Capet et al, 2014).

Scientific validation:

The methodology to define and estimate those indicators on the basis of in-situ ship-casts is presented and discussed extensively in (Capet et al, 2016). Ship-cast are processed with the DIVA detrending protocol (Capet et al, 2014) to dissociate spatial variability from seasonnal and interannual trends. This methodology is validated on the basis of synthetic datasets and by considering the better documented case of the mixed layer depth. Years with insufficient data coverage were discarded from the trend analyses. The results are concomittant with similar descriptions gathered from independent litterature as concern the oxygenation trends (eg. Konovalov et al., 2001, Pakhomova et al., 2014) and modern oxygen conditions (eg. Stanev et al, 2013). Additional details can be found in the CMEMS Ocean State Report (Capet et al., 2018).

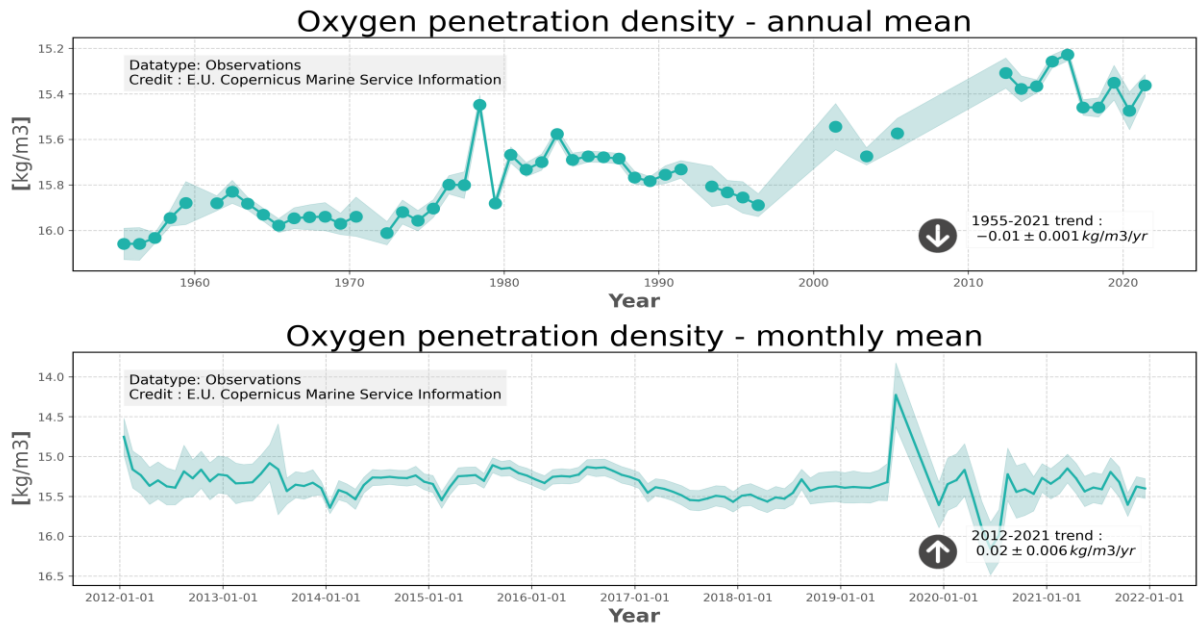

Figure 2: Potential density anomaly at the oxygen penetration depth in the open Black Sea. Upper panel: Annual trends from ship-cast (1955-2005) and autonomous platforms (Argo, 2012-2021) in-situ observations. Lower Panel: Monthly trends from Argo in-situ observations. Shaded area represent the standard error.

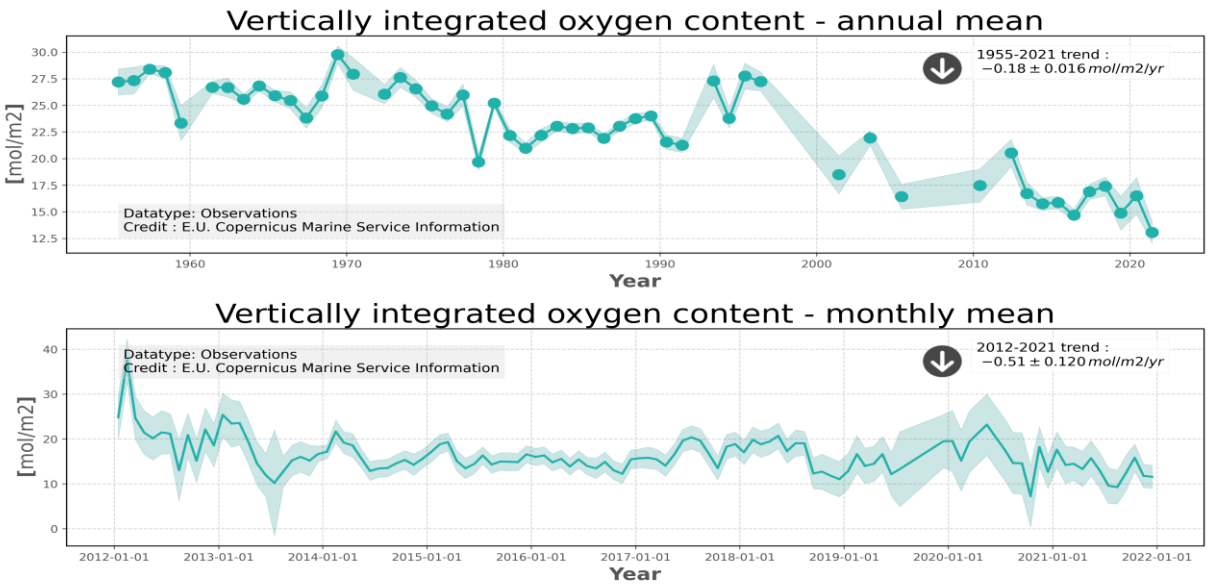

Figure 3: Oxygen inventory in the open Black Sea. Upper panel: Annual trends from ship-cast (1955-2005) and autonomous platforms (Argo, 2012-2021) in-situ observations. Lower Panel: Monthly trends from all Argo in-situ observations. Shaded area represent the standard error.

Product used for indicator evaluation:

The indicators have been evaluated on the basis of historical ship-casts (1955-2005) and more recent Argo profiles (2011-2021). Historical data were extracted from the World Ocean Database 2013 (WOD13) repository, and complemented with data from the KNORR2001, KNORR2003 and Endeavor2005 cruises. Argo data sets were extracted from the CMEMS product INSITU_BS_NRT_OBSERVATIONS_013_034.

References:

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